Shicong LIU

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Research Directions

Near-Field · Movable Antenna · XL-MIMO · Channel Estimation · AI for Wireless

Education

City University of Hong Kong — **09 2023 – 08 2027 (Expected)**
Doctor of Philosophy (Supervisor: Prof. Xianghao YU) — Hong Kong SAR, China
Electrical Engineering — 3.92/4.0

Beijing Institute of Technology — **09 2020 – 06 2023**
Master of Science (Supervisor: Prof. Zhen GAO) — Beijing, China
Information and Communication Engineering — Municipal Outstanding Graduate

Beijing Institute of Technology — **09 2016 – 06 2020**
Bachelor of Science — Beijing, China
Electronics and Information Engineering

Awards

- ICC'26 Student Travel Grant — 04 2026
- Chinese Institute of Electronics (CIE) Outstanding Master's Thesis. — 03 2026
- Exemplary Reviewer of IEEE Wireless Commun. Lett. — 02 2026
- Exemplary Reviewer of IEEE Commun. Lett. — 12 2024
- CityU Academic Excellence and QE Award — 09 2024
- Entrance Fellowship of CityU Graduate School — 09 2023
- **Hong Kong Ph.D. Fellowship Scheme (HKPFS) Awardee** — 04 2023
- 2021 **National Scholarship** (~ 0.2%) for Graduate Students — 09 2021
- 2020 **National Scholarship** (~ 0.2%) for Graduate Students — 09 2020

Research

Near-Field Communication via Curved Caustic Beams

Supervisor: Prof. Xianghao YU — 01 2026

- Leveraged EM caustics to design curved beams that focus on the legitimate user region and bypass the undesired eavesdropping region, enabling robust and secure near-field communication [J1].

Massive Movable Antennas in Near-Field Communications

Supervisor: Prof. Xianghao YU — 08 2025

- **Pioneered** the Antenna Density Function (ADF) concept, a novel analytical tool providing the first tractable model to analyze and optimize massive movable antenna systems in the near-field. [J3]
- Derived the first-known closed-form optimal antenna positions for near-field communication maximizing achievable rates [J2, J3].
- Analyzed the optimal movable antenna positions using an **Electrostatic Equilibrium** model. Derived the **closed-form solution** of optimal discrete antenna positions with asymptotic analysis [J2, C1].

Sensing Assisted Channel Estimation for Near-Field XL-MIMO

Supervisor: Prof. Xianghao YU 09 2024

- Proposed a **back-projection** based algorithm for near-field localization with significantly **reduced complexity** [J4].
- Utilized the estimated location coordinates for channel estimation/beamfocusing [C2, C3].

Learning-based Communication Signal Processing with XL-MIMO

Supervisor: Prof. Zhen Gao 06 2023

- Channel estimation and CSI feedback techniques with XL-MIMO antenna arrays [J6, C5, C7].
- Semantic communications for texts [C4].

Reconfigurable Intelligent Surface (RIS) assisted Wireless Commun.

Supervisor: Prof. Zhen Gao 09 2019

- Designed architectures and algorithms for RIS-assisted MIMO-OFDM systems.
- Proposed an OMP-based uplink channel estimation method to efficiently reconstruct the **sparse channel matrix** with low overhead. Enhanced the estimation performance with denoising neural networks [J5].
- Proposed an AMP-based user detection method to achieve grant-free multiple access [J7].

Services

- **Academic**
  - **Peer Reviewer**, IEEE ComSoc/SPS Journals and Conferences.
  - **Session Chair**, *Wireless Networks*, ICC'26, Glasgow, UK. 05 2026
  - **Session Chair**, *Near-Field, ELAA & Holo. MIMO*, ICC'26, Glasgow, UK. 05 2026
  - **TPC Member**, *Wireless Communication*, ICC'26, Glasgow, Scotland, UK. 12 2025
  - **Session Chair**, *Antenna and Smart Antenna*, GLOBECOM'24, Cape Town. 12 2024
  - **Session Chair**, *Mobile and Wireless Networks*, ICCC'23, Dalian, China. 08 2023
- **Teaching**
  - **Research Assistant** at Dept. EE, City University of Hong Kong. 08 2024 -- 10 2025
  - **Teaching Assistant**:
    - EE3009 Data Commun. and Networking, City University of Hong Kong Spring 2026
    - EE3008 Principles of Communications, City University of Hong Kong Fall 2024
    - EE3008 Principles of Communications, City University of Hong Kong Spring 2024
    - EE3008 Principles of Communications, City University of Hong Kong Fall 2023
    - Innovation and Entrepreneurship Projects, Beijing Institute of Technology Spring 2023
    - Frontiers of Communication Technology, Beijing Institute of Technology Spring 2022

Publications

Google Scholar Citations: ∼ 900

Journals

[J1] **S. Liu** and *X. Yu**, "Robust and secure near-field communication via curved caustic beam," Jan. 2026, Resubmitted to *IEEE Wireless Commun. Lett.*, Under Review.

[J2] **S. Liu**, *X. Yu**, S. Song, and K. B. Letaief, "Near-field communication with movable antennas: An electrostatic equilibrium perspective," *IEEE Trans. Wireless Commun.*, vol. 25, 2026, *Early Access.*

[J3] **S. Liu**, *X. Yu**, J. Xu, and R. Zhang, "Near-field communication with massive movable antennas: A functional perspective," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 14 455–14 470, 2026.

[J4] **S. Liu**, *X. Yu**, Z. Gao, J. Xu, D. W. K. Ng, and S. Cui, "Sensing-enhanced channel estimation for near-field XL-MIMO systems," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 3, pp. 628–643, Mar. 2025.

[J5] **S. Liu**, *Z. Gao**, J. Zhang, M. D. Renzo, and M.-S. Alouini, "Deep denoising neural network assisted compressive channel estimation for mmWave intelligent reflecting surfaces," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 9223–9228, Aug. 2020, (**ESI Highly Cited**, **Best Readings in RIS**).

[J6] *Z. Gao**†, **S. Liu**†, Y. Su, Z. Li, and D. Zheng, "Hybrid knowledge-data driven channel semantic acquisition and beamforming for cell-free massive MIMO," *IEEE J. Sel. Top. Signal Process.*, vol. 17, no. 5, pp. 964–979, Sep. 2023.

[J7] X. Zhou, K. Ying, **S. Liu**, M. Ke, *Z. Gao**, and M.-S. Alouini, "Reconfigurable intelligent surface assisted grant-free massive access," *Intell. Converg. Netw.*, vol. 3, no. 1, pp. 134–143, Mar. 2022.

Articles/Magazines

[A1] **S. Liu**, *Z. Gao**, Y. Wu, D. W. Kwan Ng, X. Gao, K.-K. Wong, S. Chatzinotas, and B. Ottersten, "LEO satellite constellations for 5G and beyond: How will they reshape vertical domains?" *IEEE Commun. Mag.*, vol. 59, no. 7, pp. 30–36, Jul. 2021.

[A2] L. Bian, X. Chang, S. Jiang, L. Yang, X. Zhan, **S. Liu**, D. Li, R. Yan, Z. Gao, and J. Zhang, "Large-scale scattering-augmented optical encryption," *Nat. Commun.*, vol. 15, no. 1, p. 9807, Dec. 2024.

Conferences

[C1] **S. Liu** and *X. Yu**, "Near-field line-of-sight communication with massive movable antennas," in *IEEE Int. Conf. Commun. (ICC'26)*, Glasgow, UK, May 2026.

[C2] **S. Liu** and *X. Yu**, "Low-complexity near-field localization with XL-MIMO sectored uniform circular arrays," in *Proc. IEEE Glob. Commun. Conf. (GLOBECOM'24)*, Cape Town, South Africa, Dec. 2024, pp. 4756–4761.

[C3] **S. Liu**, *X. Yu**, Z. Gao, and D. W. K. Ng, "DPSS-based codebook design for near-field XL-MIMO channel estimation," in *Proc. IEEE Int. Conf. Commun. (ICC'24)*, Denver, CO, USA, Aug. 2024, pp. 3864–3870.

[C4] **S. Liu**, *Z. Gao**, G. Chen, Y. Su, and L. Peng, "Transformer-based joint source channel coding for textual semantic communication," in *Proc. IEEE/CIC International Conference on Communications in China (ICCC'23)*, 2023, pp. 1–6.

[C5] **S. Liu**, *Z. Gao**, C. Hu, S. Tan, L. Fang, and L. Qiao, "Model-driven deep learning based precoding for FDD cell-free massive MIMO with imperfect CSI," in *Proc. International Wireless Communications and Mobile Computing (IWCMC'22)*, (**Student Best Paper**), Dubrovnik, Croatia, 2022, pp. 696–701.

[C6] H. Sun, **S. Liu**, X. Yu, and Y. Sun, "Flow matching-based active learning for radio map construction with low-altitude UAVs," in *IEEE Int. Conf. Acoust. Speech Signal Process (ICASSP'26)*, Barcelona, Spain, May 2026, pp. 21 406–21 410.

[C7] M. Wu, Z. Wan, Y. Wang, **S. Liu**, and *Z. Gao**, "Deep learning-based rate-splitting multiple access for massive MIMO-OFDM systems with imperfect CSIT," in *Proc. International Symposium on Wireless Communication Systems (ISWCS'22)*, 2022, pp. 1–6.

[C8] C. Zhang, H. Huang, Z. Zhang, and **S. Liu**, "Optimization of VCDTS algorithm in Connect6 game," in *Proc. Chinese Control And Decision Conference (CCDC)*, 2018, pp. 6643–6646.

Patents

[P1] Y. Wang, K. Ying, **S. Liu**, Z. Gao, D. Zheng, and J. Zhang, *A method and system for optimizing drone trajectories in IoT data collection*, CN Patent CN113382060B, Mar. 2022.